

Topic: Linear Maps and their Properties

Question 1. Determine whether each of the following maps is linear? Justify your answer.

- (a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a + b, ab, 2a - b)$.
- (b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a - b, a + b, 3^a)$.
- (c) $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0 + 1) + a_1x + a_2x^2$.
- (d) $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0 + \sin(\pi a_1)) + a_1x + a_2x^2$.
- (e) $T: \mathcal{V} \rightarrow \mathcal{P}_2(\mathbb{R})$, where $\mathcal{V}$ is the subspace of twice differentiable functions on $\mathbb{R}$, defined by $T(f(x)) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2$.

(Note: $T(f)$ is Taylor polynomial of degree two of f at the center $a = 0$.)

- (f) $T: \mathcal{V} \rightarrow \mathcal{V}$ defined by $T(f(x)) = \int_0^x e^t f(t) dt$, where $\mathcal{V}$ is the subspace of continuous functions on $[0, 1]$.

Remark: By the Fundamental Theorem of Calculus, since f is continuous on $[0, 1]$, the function $\int_0^x e^t f(t) dt$ is a continuous function (of x) on $[0, 1]$. It is also differentiable on $(0, 1)$, and $\frac{d}{dx} \left(\int_0^x e^t f(t) dt \right) = e^x f(x)$.

Question 2. Let R_θ be the counter-clockwise rotation on $\mathbb{R}^2$ by an angle θ about the origin.

- (a) Show that

$$R_\theta \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

for every $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

- (b) Hence, deduce that R_θ is linear.
- (c) Find the image of the point $(1, 2)$ under the counter-clockwise rotation by $\pi/3$ radians on about the origin.

Question 3.

- (a) Prove that there exists a linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(1, 1) = (1, 0, 2)$ and $T(2, 3) = (1, -1, 4)$. What is $T(8, 11)$?
- (b) Is there a linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(1, 0, 3) = (1, 1)$ and $T(-2, 0, -6) = (2, 1)$?

Question 4. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map, where $\mathcal{V}$ and $\mathcal{W}$ are real vector spaces. Suppose $S_1 = \{v_1, v_2, \dots, v_k\}$ is a subset of $\mathcal{V}$ and $S_2 = \{w_1, w_2, \dots, w_k\}$ is a subset of $\mathcal{W}$ such that $T(v_i) = w_i$, for each $i = 1, 2, \dots, k$.

(Remark: The vector $v_i \in \mathcal{V}$ such that $T(v_i) = w_i$ is called a pre-image of w_i under T .)

Determine whether each of the following statements is true or false. Justify your answers.

- (a) If S_1 is linearly independent, then S_2 is linearly independent.
- (b) If S_2 is linearly independent, then S_1 is linearly independent.

Question 5. Suppose $T, S: \mathcal{V} \rightarrow \mathcal{W}$ are linear maps between vector spaces $\mathcal{V}$ and $\mathcal{W}$. Prove that the following maps are linear.

- (a) $T + S: \mathcal{V} \rightarrow \mathcal{W}$ where $(T + S)(v) = T(v) + S(v)$ for all $v \in \mathcal{V}$.
- (b) $\mu T: \mathcal{V} \rightarrow \mathcal{W}$ where μ is a scalar and $(\mu T)(v) = \mu T(v)$ for all $v \in \mathcal{V}$.

Question 6. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map, where $\mathcal{V}$ and $\mathcal{W}$ are vector spaces over $\mathbb{R}$.

- (a) Suppose $\mathcal{V}_1$ is a subspace of $\mathcal{V}$. Let $T(\mathcal{V}_1)$ be the subset of $\mathcal{W}$ containing all images of vectors in $\mathcal{V}_1$ under T , i.e., $T(\mathcal{V}_1) = \{T(v) : v \in \mathcal{V}_1\}$. Prove that $T(\mathcal{V}_1)$ is a subspace of $\mathcal{W}$.
- (b) Suppose $\mathcal{W}_1$ is a subspace of $\mathcal{W}$. Let $T^{-1}(\mathcal{W}_1)$ be the subset of $\mathcal{V}$ containing all pre-images of vectors in $\mathcal{W}_1$, i.e., $T^{-1}(\mathcal{W}_1) = \{v \in \mathcal{V} : T(v) \in \mathcal{W}_1\}$. Is $T^{-1}(\mathcal{W}_1)$ a subspace of $\mathcal{V}$? (Caution: T may not be bijective and so the inverse $T^{-1}(\{v\})$ may be empty or contain more than one element.)

Question 7. Let $T: \mathcal{M}_{n \times n}(\mathbb{R}) \rightarrow \mathcal{M}_{n \times n}(\mathbb{R})$ and $U: \mathcal{M}_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$, defined by

$$T(A) = A + A^T, \quad U(A) = \text{tr}(A).$$

Determine UT . (Simplify it!) Here, if $A = (a_{ij})_{1 \leq i, j \leq n}$, then $\text{tr}(A) = \sum_{i=1}^n a_{ii}$. That is, $\text{tr}(A)$ is the sum of all diagonal entries of A .

Topic: Null Space, Range

Question 8. Consider $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} a+b \\ 0 \\ 2a-b \end{bmatrix}$.

- (a) Find a matrix A such that $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = A \begin{bmatrix} a \\ b \end{bmatrix}$. Hence, deduce that T is a linear map.
- (b) Find bases for both null space $\text{null}(T)$ and range $\text{range}(T)$.

Question 9. For each of the following linear maps, determine if T is (i) injective (ii) surjective (iii) invertible.

(a) $T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ defined by $T(f(x)) = f'(x)$.

(b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear with $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

(c) $T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ defined by $T(f)(x) = \int_0^x f(t)dt$. Specifically, T maps any $f \in \mathcal{P}(\mathbb{R})$ to its antiderivative.

Question 10. Consider the linear map $T: \mathcal{M}_{2 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{2 \times 2}(\mathbb{R})$ defined by

$$T\left(\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}\right) = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix}.$$

(a) Find a basis for $\text{range}(T)$. Is T surjective?

(b) Determine $\text{nullity}(T)$. Is T injective?

(c) Find a basis for $\text{null}(T)$.